Sahaj Bhadja

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Summary

Research-driven solutions specialist translating **scientific computing, AI/ML, Linux, cloud infrastructure, data storage, and security** requirements into scalable architectures, technical recommendations, and measurable customer outcomes.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Translated data-intensive research requirements into scalable FastAPI, PostgreSQL, SQL Server, and cloud-storage workflows processing 5,000+ monthly jobs, increasing concurrent throughput 3x while reliably sustaining 99%+ production reliability.
- Designed multi-tenant data architecture for 500+ users, reliably reducing dashboard latency from 1.2 seconds to 350 milliseconds through storage modeling, indexing, caching, workload analysis, and measured optimization.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed AI/ML workflows using Python, PyTorch, multimodal models, AWS RDS, and REST APIs, supporting 150+ instructors and efficiently processing 10,000+ daily requests with stable performance at scale.
- Analyzed model usage, token consumption, workload failures, and estimated costs across GPT-4o, Claude, and Qwen, implementing diagnostics and fallback strategies while reducing feedback turnaround time by 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built scientific computing pipelines processing 100,000+ trajectory records, reliably improving long-horizon planning 18% through Python, PyTorch, simulation, model training, performance analysis, and reproducible experimentation at production scale.
- Translated complex research findings for senior researchers, clearly documenting system architecture, experimental tradeoffs, and technical recommendations while improving closed-loop stability 15% across ego, road, and nearby-agent interactions.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform data infrastructure across Linux, Windows, and macOS using Java, Python, Docker, AWS Glue, and S3, reliably increasing pipeline efficiency 68% while reducing storage costs 15%.
- Collaborated with Microsoft and GitHub stakeholders from requirements through deployment, automating 20+ tests, technical reports, coverage summaries, and failure logs while reducing API response times by 30%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Mapped 12+ integration points and clearly presented end-to-end architecture recommendations to the client, translating business requirements into cross-platform booking, payment, notification, and real-time status capabilities at scale.
- Delivered Flask, Firebase, and cloud-based solutions across 50,000+ records, supporting client reporting and analytics while achieving sub-200-millisecond responses and reducing manual refund handling by 70% for customers.

Skills

Scientific Computing: Python, Linux, Bash, PyTorch, NumPy, Pandas, simulation, model training, data analysis
AI/Infrastructure: AI/ML, LLMs, GPU workloads, cloud computing, Docker, Kubernetes, AWS, Azure, GCP
Data/Security: PostgreSQL, SQL Server, MongoDB, S3, cloud storage, ETL, firewalls, encryption, access controls
Solutions: Requirements analysis, architecture design, technical research, presentations, documentation, CRM, cost analysis

Projects

Big-Brain Live | *AI Research Computing Architecture*
- Built AI-powered retrieval over 50,000+ documents using PostgreSQL, GCP, LLaMA-2, FAISS, Pinecone, and REST APIs, delivering sub-second search while reducing retrieval latency 40% through scalable infrastructure optimization.

CPU-scheduling Simulator Live | *Computational Resource Optimization*
- Implemented six C++ scheduling algorithms with automated workload metrics and Gantt visualization, evaluating 1,000+ process combinations while reducing average turnaround time 25% and debugging time 20% through comparative analysis.

CaptionCraft - Image Caption Generator Live | *GPU-Accelerated AI/ML Research*
- Benchmarked BLIP, GIT, ViT-GPT-2, and custom architectures across 10,000 images and 100,000 captions, evaluating computational approaches and improving measured accuracy from 81% to 96% through structured experimentation.